Topology Exam 1 Cheatsheet

Raymond Langehennig
April 28, 2026

Topological spaces

topology A set T of subsets of X such that

1. $X \in T$
2. $\varnothing \in T$
3. T is closed under *arbitrary unions* and *finite intersections*

basis A collection $\mathcal{B}$ of subsets of X such that

1. $\forall x \in X, \exists B \in \mathcal{B} : x \in B$
2. $\forall B_1, B_2 \in \mathcal{B}, x \in B_1 \cap B_2 \implies \exists B_3 \in \mathcal{B} : x \in B_3 \subseteq B_1 \cap B_2$

The set $T_{\mathcal{B}} = \{\bigcup \mathcal{F} \mid \mathcal{F} \subset \mathcal{B}\}$ is the topology generated by $\mathcal{B}$.

fineness T is finer than $S \iff S \subset T$

subspace (Y, T_Y) where $Y \subset X$ and $T_Y = \{Y \cap U \mid U \in T\}$

closure $\overline{A} = \mathrm{Cl}(A) = \bigcap \{U \mid \complement_X U \in T \wedge A \subset U\}$ is the smallest closed set containing A

interior $\mathrm{Int}(A) = \bigcup \{U \mid U \in T \wedge U \subset A\}$ is the largest open set contained in A

boundary point $x \in \mathrm{Cl}(A) \setminus \mathrm{Int}(A)$

limit point L is a limit point in a subspace $X \iff$ $\forall$open neighborhood U of $L, U \cap X \neq \varnothing$

limit of sequence x for a sequence $(x_n)_{n\in\mathbb{N}}$ of points in X $\iff \forall U \in T, x \in U \implies \exists N \in \mathbb{N} : \forall n \geq N, x_n \in U$. Equivalently though informally, there comes a point where the terms of the sequence remain within a neighborhood of x.

convergence existence of the limit of a sequence

pointwise convergence for a sequence of functions $(f_i)_{i\in\mathbb{N}}$, if $(f_i(x))_{i\in\mathbb{N}}$ converges for all x in the domain of f_i.

uniform convergence $\forall \varepsilon > 0, \exists N \in \mathbb{N} : \forall n > N, \forall x \in X, d(f_n(x), f(x)) < \varepsilon$. converges to f

Properties

connected if $\nexists$ open sets $U, V : U \cup V = X \wedge U \cap V = \varnothing$, or equivalently X and $\varnothing$ are the only clopen sets

path-connected if $\forall p, q \in X, \exists$ continuous $(\phi : [0,1] \to X) : \phi(0) = p \wedge \phi(1) = q$. Implies connectedness.

compact given that a *cover* of $Y \subseteq X$ is a collection $\mathcal{C}$ of subsets of X such that $\bigcup \mathcal{C} \supseteq Y$, and an *open cover* is a cover whose elements are open, Y is compact if every open cover of Y includes a finite open cover of Y.

Special cases

trivial $T = \{\varnothing, X\}$

discrete $T = \mathcal{P}(X)$

T_2 $\iff \forall x_1, x_2 \in X, \exists$ neighborhood U_1 of $x_1 \exists$ neighborhood U_2 of $x_2 : U_1 \cap U_2 = \varnothing$
$\iff$ limits of sequences are unique

metric for metric space (X, d), the topology generated by $\{B_d(p, r) \mid p \in X \wedge r \in \mathbb{R}\}$.

order for $(X, <)$, the topology generated by the collection $\mathcal{B}$ of the following subsets $\forall a, b \in X$.

- (a, b)
- $[\min(X), b)$
- $(a, \max(X)]$

product for a collection of topological spaces of the form (X, T_X), the collection of their underlying sets $\mathcal{X}$, the topology generated by $\{\prod_{i\in I} U_i \mid U \in T_i\}$.

on $\mathbb{R}, \forall x, y \in \mathbb{R}$:

standard the metric topology induced by $d(x, y) = \sqrt{x^2 + y^2}$

lower generated by the basis of $[a, b)$

upper generated by the basis of $(a, b]$

k-topology generated by the basis of $(a, b) \setminus \{\frac{1}{z} \mid z \in \mathbb{Z}_+\}$

Maps

continuous a map $f : X \to Y$ between topological spaces (X, T) and (Y, S) if $\forall V \in S, f^{-1}(V) \in T$. Continuous maps preserve

- connectedness
- path-connectedness
- compactness

homeomorphism a bijection $f : X \to Y$ if both f and f^{-1} are continuous

closed map $f : X \to Y$ if $\forall A \subset X, A$ closed $\implies f(A)$ closed

Metric spaces

metric on a set X, the function $d : X \times X \to \mathbb{R}$ such that, $\forall x, y, z \in X$,

1. $d(x, y) \geq 0$
2. $d(x, y) = 0 \iff x = y$
3. $d(x, y) = d(y, x)$
4. $d(x, z) \leq d(x, y) + d(y, z)$

A metric *induces* the topology generated by the basis of all open balls using d.

For $A \subset X$,

- $d(x, A) := \inf \{d(x, a) \mid a \in A\}$
- $\mathrm{diam}\, A := \sup \{d(a_1, a_2) \mid a_1, a_2 \in A\}$

ε-neighborhood the set

$$\begin{aligned} U(A, \varepsilon) &:= \{x \mid d(x, A) < \varepsilon\} \\ &= \bigcup_{a\in A} B_d(a, \varepsilon) \end{aligned}$$

Theorems

Uniform limit For a sequence of functions (f_n) from a topological space to a metric space, uniform convergence to $f \implies f$ is continuous.

Extreme value

Lebesgue covering For a compact metric space X with open cover $\mathcal{C}$, there is some number $\delta > 0$ such that every subset of X with diameter less than δ is contained in some member of $\mathcal{C}$.

Uniform continuity $f : X \to Y$ is a continuous map from the compact metric space (X, d_X) to the metric space $(Y, d_Y) \implies f$ is uniformly continuous.

Pf. $\mathcal{B} := \{B_{d_Y}(y, \varepsilon/2) \mid y \in Y\}$ for some $\varepsilon > 0$ is an open covering of Y. Then $\mathcal{A} := \left\{f^{-1}(B) \mid B \in \mathcal{B}\right\}$ is an open covering of X, with Lebesgue number δ. Then

$$\begin{aligned}(x_1, x_2 \in X) \wedge (d_X(x_1, x_2) < \delta) \\ &\implies \operatorname{diam}\{x_1, x_2\} < \delta \\ &\implies \{f(x_1), f(x_2)\} \in B \in \mathcal{B} \\ &\implies d_Y(f(x_1), f(x_2)) < \varepsilon\end{aligned}$$

Miscellanea

total order a relation $<$ on a set S such that $\forall a, b, c \in S$:

- $a < a$
- $a < b \wedge b < c \implies a < c$
- $a < b \wedge b < a \implies a = b$

linear continuum a totally ordered set L such that:

- every subset bounded above has a least upper bound
- $\forall a, b \in L, a < b \implies \exists c \in L : a < c < b$ (density)

path between points p and q in X, a continuous map $\phi : [0, 1] \to X$ such that $\phi(0) = p$ and $\phi(1) = q$